

2026 Post-ICML Workshop

□ Introduction

The 2026 ICML (International Conference on Machine Learning) conference, a leading top AI academic conference, will be held in Seoul during July 6-11. This workshop aims to invite researchers from the conference attendees to share their latest research findings and trends, centering on AI safety, security of mobile on-device AI, data-centric graph learning, multimodal AI, and LLM uncertainty and evaluation. Additionally, some speakers will cover more on the behind-the-story of papers: motivation and impetus for a paper's initiation, the process of problem formulation and corresponding idea generation, challenges and responses during the process, the review and rebuttal process, the utilization of collaboration network, hard-learned lessons, etc. Early career researchers including post-graduate students and post-doctoral fellows can learn best practices in research methodologies, collaboration networking and career management.

□ Overview

- o Date: July 10th (Friday), 2026
- o Venue: Gachon University, AI Hall, Room 100
(<https://maps.app.goo.gl/fnUXTbBcbcuPwa3Z6>)

□ Schedule

- o The presentation time will be managed flexibly according to the presentation situation.

Time (KST)	Title	Speaker
10:00 – 10:10	Introduction of Distinguished Speakers, Guests, and Attendees	Dr. Seong Oun Hwang
10:10 – 11:00	Building Digital Trust: Decentralized Identity and AI Safety in the Digital Era	Prof. Lam Kwok Yan Dr. Linru Zhang Dr. Yongsen Zheng
11:00 – 11:50	The Security Landscape of Mobile On-device AI: From Local Models to Real-World Threats	Dr. Yujin Huang
11:50 –	Lunch Break	

12:40	(A light lunch will be provided by the organizer.)	
12:40 – 13:30	Automated Graph Machine Learning: Toward a Data-centric Framework	Prof. Xin Zheng
13:30 – 14:20	Recent Trends in Multimodal AI at ICML 2026	Linta Iqbal
14:20 – 15:10	SAGE: A Search-AuGmented Evaluation of Large Language Models on Free-Form QA SCOPE: Selective Conformal Optimized Pairwise LLM Judging	Sher Badshah
15:10 – 15:30	Q&A and Discussion	Prof. Seong Oun Hwang

Abstract

Speaker	Abstract
Prof. Lam Kwok Yan (Vice President), Dr. Linru Zhang, Dr. Yongsen Zheng (Nanyang Technological University, Singapore)	This talk explores how digital trust can be built in the digital era through two foundational pillars: decentralized identity and AI safety. The talk discusses that decentralized identity and rigorous AI safety testing together form a critical foundation for secure, privacy-preserving, and accountable digital ecosystems.
Dr. Yujin Hwang (Post-doctoral Fellow, University of Melbourne, Australia)	Mobile on-device AI (MoAI) systems that integrate locally deployed AI models with conventional mobile software components are emerging as a key paradigm for delivering intelligent functionality directly on end-user devices. By moving inference from remote cloud services to the local mobile environment, such systems enable privacy-preserving, low-latency, and offline-capable AI functionality, yet introduce new security risks arising from the local storage of AI models. This paper presents the first comprehensive systematization of knowledge on MoAI security, covering security pillars, attack landscape, and defense landscape of MoAI systems. We further identify unresolved gaps in current attack and defense research and point to promising directions for future research in this emerging area. Our work establishes the first systematic framework for understanding the attack and defense landscapes of MoAI systems, serving as a foundation for building secure MoAI systems and advancing research in this critical domain.
Prof. Xin Zheng (RMIT University, Australia)	Automated graph machine learning is emerging as a powerful paradigm for building efficient and deployable models across a wide range of real-world applications, from scientific discovery to recommender systems. In this talk, Dr. Xin Zheng presents a data-centric perspective on automated graph learning, focusing on the full pipeline—from graph data-level scalability to graph model-level deployment. The talk introduces a structured framework that addresses three key dimensions (1) Scalability, through data-driven graph condensation techniques (2) Automation, via neural architecture search guided by graph properties and learning tasks; and (3) Deployability, through test-time evaluation and adaptation mechanisms for practical model deployment. The

	presentation offers a unified perspective on efficient and deployable automated and data-centric graph learning.
Linta Iqbal (Post-graduate Researcher, Gachon University, Korea)	This presentation will offer a preview of the key trends in multimodal AI expected to emerge from ICML 2026, based on the conference's accepted papers, invited talks, and workshop themes. The focus will be on identifying shifts towards current AI practices. Particular attention will be given to how multimodal systems are evolving to handle more complex tasks and real-world interactions. Finally, key takeaways and their potential implications for our ongoing work will be discussed.
Sher Badshah (Post-graduate Researcher, Dalhousie University, Canada)	SAGE: A Search-AuGmented Evaluation of Large Language Models on Free-Form QA As Large Language Models (LLMs) become increasingly used for question-answering (QA), relying on static, pre-annotated references for evaluation poses significant challenges in cost, scalability, and completeness. Meanwhile, using LLMs themselves as evaluators without external grounding remains unreliable for objective tasks, as they systematically over-accept incorrect answers, fabricate supporting rationales, and degrade sharply on questions that fall outside their training data. We propose Search-AuGmented Evaluation (SAGE), a framework to assess LLM outputs without fixed ground-truth answers. Unlike conventional metrics that compare to static references or depend solely on LLM-as-a-judge knowledge, SAGE acts as an agent that actively retrieves and synthesizes external evidence. It iteratively generates web queries, collects information, summarizes findings, and refines subsequent searches through reflection. By reducing dependence on static reference-driven evaluation protocols, SAGE offers a scalable and adaptive alternative for evaluating the factuality of LLMs. Experimental results on multiple free-form QA benchmarks show that SAGE achieves substantial to perfect agreement with human evaluations. SCOPE: Selective Conformal Optimized Pairwise LLM Judging Large language models (LLMs) are increasingly used as scalable judges in pairwise evaluation, but they remain prone to miscalibration and biases. We propose SCOPE (Selective Conformal Optimized Pairwise Evaluation), a framework that calibrates an acceptance threshold so that, under exchangeability, the error rate among non-abstained judgments is at most a user-specified level α. To supply SCOPE with a bias-neutral uncertainty signal, we introduce Bidirectional Preference Entropy (BPE), which queries the judge under both response positions and converts the order-averaged preference probability into an entropy-based score. Across various pairwise judging benchmarks, BPE outperforms standard confidence proxies in calibration and discrimination, while SCOPE consistently satisfies the target risk bound (empirical FDR ≈ 0.097–0.099 at $\alpha = 0.10$) and retains substantial coverage. Compared to vanilla baselines, SCOPE accepts up to 2.4$\times$ more judgments under the same risk constraint, demonstrating that BPE enables reliable and high-coverage LLM-based evaluation.

